

## 07.03 Adding and Subtracting Rational Expressions

Common problems and solution methods

### Rule guide: recognize the task first

| Recognize                                                    | Apply these rules                                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Rational Expressions And Equations                           | State denominator restrictions before simplifying or solving.                                                  |
| The expression contains the chapter's key notation or model. | Write the known quantities, choose the matching formula, calculate carefully, and check the result in context. |

### Example 1: Identify the model

#### Problem

$$\frac{x^2 - 9}{x^2 + 3x}$$

#### How to solve it

Write the known quantities and restrictions.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

Check the result against the original statement.

### Example 2: Apply the key rule

#### Problem

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

#### How to solve it

Write the known quantities and restrictions.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

Check the result against the original statement.

### Example 3: Check the result

#### Problem

$$\frac{x+1}{x^2-4}$$

#### How to solve it

Write the known quantities and restrictions.

# LCD clears denominators

Check the result against the original statement.